

Assessing Autonomic Function by Analysis of Heart Rate Recovery from Exercise in Healthy Subjects

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Although delayed recovery of heart rate (HR) after exercise indicates poor prognosis, the relative role of parasympathetic reactivation versus sympathetic withdrawal in controlling exercise HR recovery remains controversial. Quantifying HR recovery is difficult because the rate of recovery varies with exercise level. This study develops a model of HR recovery applicable to multiple exercise levels simultaneously. Using the Levenberg-Marquardt method for nonlinear models, HR curves for 11 healthy volunteers recovering from 4 different levels of exercise were fit to equations incorporating 1 first-order time constant for parasympathetic reactivation and 1 for sympathetic withdrawal. Results provided time constants for parasympathetic reactivation of 44 ± 37 seconds and for sympathetic withdrawal of 65 ± 56 sec-

onds. The model fit the HR recovery curves very closely, explaining $99.7 \pm 0.1\%$ of the variance in the data. In conclusion, this study presents a unique method for quantitatively testing theories on the relative roles of sympathetic withdrawal and parasympathetic reactivation during recovery from exercise. It provides indexes of dynamic sympathetic and parasympathetic functions, with the parasympathetic system having a faster response time. It supports theories of coordinated interaction of parasympathetic reactivation and sympathetic withdrawal during exercise recovery and does not support using simple measures of exercise HR recovery as indexes of vagal function alone. ©2004 by Excerpta Medica, Inc.

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Hear rate (HR) recovery after exercise is mediated by a combination of sympathetic withdrawal and parasympathetic reactivation. Based on studies of normal patients with and without autonomic blockade, Savin et al¹ postulated that sympathetic withdrawal contributes more to HR recovery soon after peak exercise, with parasympathetic reactivation playing a greater role later in recovery. Because the parasympathetic system reacts faster than the sympathetic, this theory would also explain the more rapid recovery from lower exercise levels, as there is less activation of the sympathetic system at submaximal exercise. Other investigators have proposed a more dominant role of parasympathetic reactivation in early postexercise HR recovery,^{2,3} but none of the postulates has been applied quantitatively to describe exercise HR recovery. This study was designed to define the kinetics of postexercise HR recovery in terms of sympathetic and parasympathetic activity by using quantitative methods that incorporate the hypothesis of Savin et al.¹

METHODS

This study was approved by the human studies subcommittee of the Minneapolis VA Medical Center and conforms to the recommendations of the Decla-

ration of Helsinki. Eleven healthy volunteers (4 men and 7 women; age range 24 to 56 years; mean 43 ± 10) exercised on a treadmill using a gradually progressive workload starting at 0 grade and 1 mile/hour and increasing by 1% grade and 0.2 to 0.3 miles/hour every minute. Exercise was performed 4 times, with a minimum of 24 hours between each test. The first exercise was to exhaustion, and successive exercises were performed in random order to levels at which the HR increase was 80%, 60%, or 40% of maximum. HR analysis started immediately after cessation of exercise, with the subject standing quietly at rest, and continued for 5 minutes. The electrocardiogram was recorded continuously on a Marquette 8500 Holter Recorder and edited on a Marquette MARS 8000 ambulatory monitoring system (Marquette, Milwaukee, Wisconsin). The annotated record was transferred to a spreadsheet where premature beats were removed and beat-to-beat HR was generated from the RR intervals.

Data are presented as mean $\pm$ SD, and differences were considered statistically significant at $p < 0.05$. Student's paired *t* test was used to compare results when initial assumptions varied. Analysis of variance for repeated measures on the same elements was used for comparisons between multiple sets of exercise levels. How well the model fit the data was assessed by using the goodness-of-fit R^2 model, expressed as percent total variance explained by the model.

Mathematical analysis: The heart has an intrinsic HR⁴ that can be modified by sympathetic and parasympathetic activities. Mathematically:

$$HR = HR_i \times HR_p \times HR_s$$

where HR_i is intrinsic heart rate, HR_p is parasympa-

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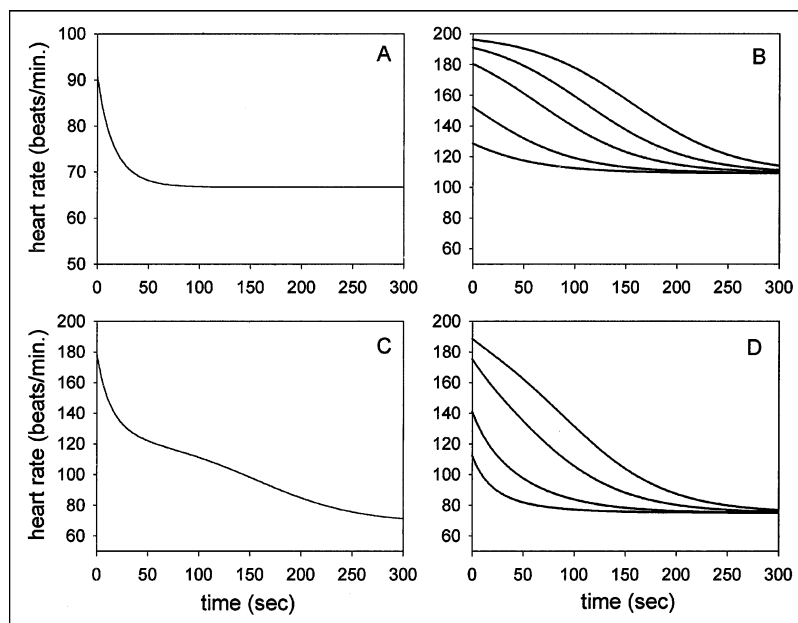


FIGURE 1. (A) Theoretical time course of HR response to parasympathetic reactivation assuming constant sympathetic activity. (B) Sympathetic withdrawal assuming constant parasympathetic activity. (C) Combined parasympathetic reactivation and sympathetic withdrawal assuming no interaction. (D) Combined parasympathetic reactivation and sympathetic withdrawal assuming central integration of the 2 responses. Note the difference in vertical (HR) scale for (A) versus (B), (C), and (D).

thetic influence, and HRs is sympathetic influence. Note that HRp must be ≤ 1.0 and approaches 0 as parasympathetic activity increases. A reasonable mathematical model is:

$$HR_p = 1/(1 + P)$$

where P is parasympathetic activity (in arbitrary units, such as vagal nerve activity in pulses per second, number of cholinergic receptors occupied, and so forth). In contrast, HRs must be ≥ 1.0 and reach an asymptote as maximal HR is achieved. Mathematically:

$$HR_s = 1 + (M \times S)/(M + S)$$

where M determines maximum HR and S represents sympathetic activity in arbitrary units. Combining these terms produces:

$$HR = HR_i \times [1/(1 + P)] \times [1 + (M \times S)/(M + S)].$$

At peak exercise, P would be minimal (parasympathetic withdrawal) and S maximal (sympathetic activation). Thus, at maximum HR (HRm), P is close to 0 and

$$HR_p = 1/(1 + P) \text{ becomes } HR_p = 1/(1 + 0) = 1.$$

Also, at maximum HR, S is very large relative to M, so $(M \times S)/(M + S)$ can be approximated by M. Thus, at maximum HR (HRm), the equation

$$HR = HR_i \times [1/(1 + P)] \times [1 + (M \times S)/(M + S)]$$

becomes

$$HR_m = HR_i \times (1 + M).$$

If the 2 systems follow first-order kinetics, then for parasympathetic reactivation:

$$P = P_o + (P_r - P_o) \times (1 - e^{-t/T_p})$$

where t represents time, P_o indicates parasympathetic activity at onset of recovery (cessation of exercise), P_r is resting (final) parasympathetic tone, and T_p is the rate constant for parasympathetic reactivation. For sympathetic withdrawal:

$$S = S_r + (S_o - S_r) \times e^{-t/T_s}$$

where S_r determines resting sympathetic tone, S_o represents sympathetic activity at cessation of exercise, and T_s is the rate constant for sympathetic withdrawal.

A theoretical curve for the effects of parasympathetic reactivation on HR (holding sympathetic activity constant) and the effect of sympathetic withdrawal (holding parasympathetic activity constant) are shown in Figure 1, which includes curves simulating several levels of exercise.

Using these equations, HR recovery from maximal exercise would appear as shown in Figure 1. It is evident that such a curve does not mimic what happens during normal exercise recovery. It is likely that central integration of autonomic function delays parasympathetic reactivation until sympathetic withdrawal has begun to prevent peripheral antagonism of the 2 systems. A simple postulate would be that parasympathetic activity is decreased in proportion to the degree of sympathetic activation. Mathematically, P in the equations is decreased in proportion to how high the HR is above the intrinsic HR, that is, multiplying P by:

$$(HR_m - HR)/(HR_m - HR_i).$$

A graph of HR recovery from several different levels of exercise presented in Figure 1 demonstrates this interactive effect.

Preliminary inspection of the data showed that HR returned to baseline from the 2 lower levels of exercise but did not completely recover within 5 minutes after the higher level of exercise. To allow for this delayed return to baseline after vigorous exercise, we allowed S_r to change with exercise level, that is, reset sympathetic tone.

To test the model, these equations were simultaneously fit to the 4 HR recovery curves for each subject by using the Levenberg-Marquardt method for nonlinear models, with calculations performed in Mathematica 4.1 (Wolfram Research, Inc., Champaign, Illinois). To allow S_o and S_r to change with exercise level, they were written as polynomial func-

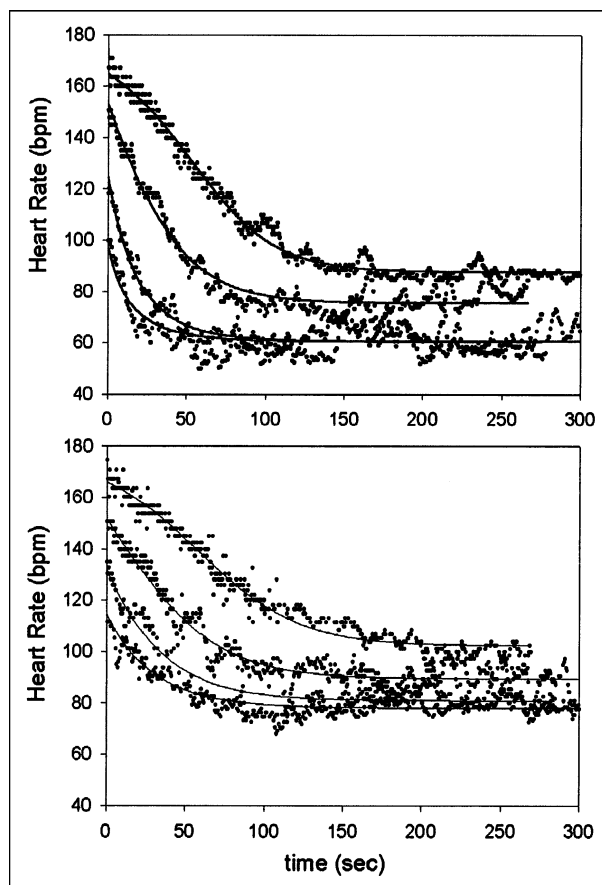


FIGURE 2. Examples of HR recovery curves with the raw data (dotted lines) and fit curves (solid lines) for 2 subjects exercising to 4 different levels.

tions of j , where $j = 0, 1, 2, 3$, corresponds to recoveries beginning from 100%, 80%, 60%, and 40% of maximal HR, respectively. The exercise responses for each patient were also modeled with subsets of the 4 exercise levels. When the model included all 4 exercise levels, the cubic polynomials $S_o = a j^3 + b j^2 + c j + d$ and $S_r = e j^3 + f j^2 + g j + h$ were used, which together added 8 parameters to the nonlinear search space. Three exercise levels were modeled by fixing $a = e = 0$, and 2 levels were modeled by fixing $c = g = 0$.

RESULTS

The model was applied to fit the exercise data for 10 of the 11 subjects. One subject was excluded because of an atypical exercise response, in which HR after maximal exercise returned to a level lower than that after 80% of maximal. Examples of the data for 2 of the subjects and the predicted curves obtained by the model are presented in Figure 2. As demonstrated in Figure 3, T_p was less than T_s for all but 1 of the 10 subjects.

No constraints were placed on any of the parameters in the model. To simplify the model, we hypothesized that at peak exercise parasympathetic activity is negligible. This hypothesis is supported by Robinson et al,⁵ who found no change in maximal exercise HR

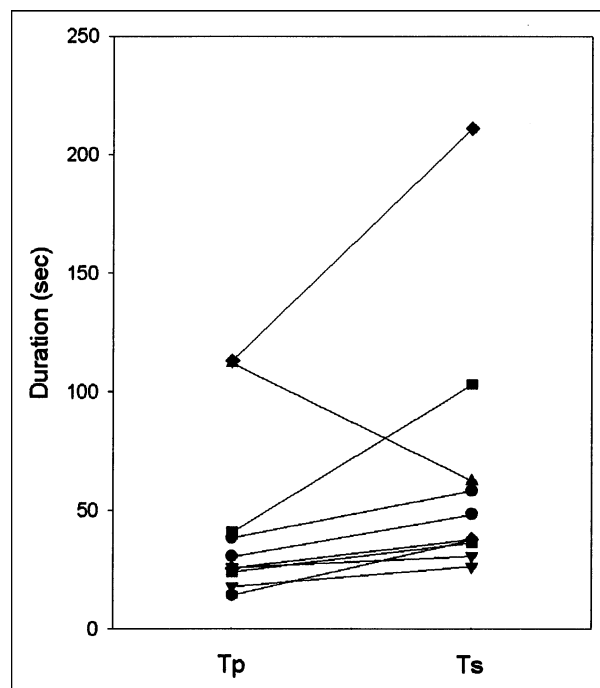


FIGURE 3. Time constants for parasympathetic reactivation (T_p) and sympathetic withdrawal (T_s) for 10 subjects at 4 exercise levels for each.

with or without atropine. At submaximal levels of exercise, Maciel et al⁶ found progressively less effect of atropine on HR as exercise level and duration increased. To test the hypothesis that parasympathetic effects may be negligible at moderate or higher levels of exercise, we set $P_o = 0$ and repeated the modeling. Results showed no significant difference in T_p (44.4 ± 36.9 vs 43.9 ± 36.0 , $p = 0.33$) or T_s (65.3 ± 55.9 vs 64.3 ± 53.0 , $p = 0.33$).

Eliminating P_o allows further simplification of the model. The difference between HR_r and HR_m is the sum of that due to parasympathetic withdrawal and that due to sympathetic activation. Because parasympathetic withdrawal cannot result in an HR higher than the HR_i , the net contribution of the parasympathetic nervous system to the maximal change in HR would be $HR_i - HR_i \times 1/(1 + Pr)$. Expressing the total change in HR from rest to maximal exercise as the sum of parasympathetic withdrawal plus sympathetic activation produces:

$$HR_m - HR_r = [HR_i - HR_i \times 1/(1 + Pr)] + (HR_m - HR_i)$$

which simplifies to:

$$HR_i = HR_r \times (1 + Pr).$$

Maximum HR can be obtained directly from the data at peak exercise and resting HR by averaging HR after low-level exercise. Substituting $HR_r \times (1 + Pr)$ for HR_i in the equations and setting $P_o = 0$ eliminate 2 parameters from the model. Doing this does not change the results significantly for T_p (43.9 ± 36.0 vs

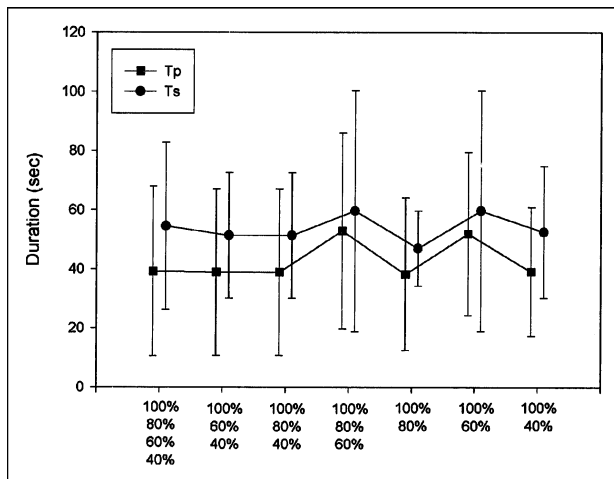


FIGURE 4. Time constants for parasympathetic reactivation (Tp) and sympathetic withdrawal (Ts) for 10 subjects when various levels of exercise are included.

39.3 ± 28.7 , $p = 0.34$) or Ts (64.3 ± 53.0 vs 54.5 ± 28.3 , $p = 0.34$). Assessing the goodness of fit for this final model indicates that the model explained $>99\%$ of the variance for every subject (mean $99.7 \pm 0.1\%$). If the deviation from the model is determined separately for each exercise curve, the data from the higher level of exercise fit better than those from lower levels. At maximum exercise, the model explained $97.6 \pm 0.7\%$ of the variance, with $92.0 \pm 4.8\%$, $78.1 \pm 21.2\%$, and $52.6 \pm 23.2\%$ at 80%, 60%, and 40% of maximum HR, respectively.

Using the simplified equations, we can test the hypothesis that all 4 levels of exercise are not necessary to adequately define a patient's dynamic parasympathetic and sympathetic activities. To do this, we determined Tp and Ts using all 4 levels of exercise and compared the results with Tp and Ts obtained after sequentially omitting various exercise levels (but always including the maximum). Analysis of variance for repeated measures showed no statistical difference in Tp or Ts between the various combinations of exercise levels (Figure 4).

DISCUSSION

First-order exponential decay has been used to model HR after exercise,^{2,7-9} with the implication that the time constant would provide a useful index of HR recovery. We found multiple problems with first-order decay,¹⁰ whereas Broman and Wigertz¹¹ found the best fit with a second-order equation, and Crouse et al¹² used a polynomial function. In a simplified approach, Cole et al chose percent recovery 1 or 2 minutes after maximal¹³ or submaximal¹⁴ exercise, with¹³ or without¹⁴ a cool-down period, as their prognostic index. None of these methods can sort out the relative contribution of parasympathetic versus sympathetic effects.

The limitations of previous models prompted our new approach, which uses physiologic correlates capable of explaining the response to multiple levels of

exercise. The final equations express HR as a function of intrinsic HR modified by parasympathetic and sympathetic influences. Maximum and resting HRs are obtained directly from the data, parasympathetic activity is considered negligible at peak exercise, and the primary process for parasympathetic reactivation and sympathetic withdrawal after exercise are exponential. The parasympathetic system is inhibited in direct proportion to sympathetic activation (central neural control).

Using a parasympathetic and sympathetic multiplier of the intrinsic HR is similar to the approach used by Rosenblueth and Simeone¹⁵ in studies of neural control in cats and by Katona et al¹⁶ in studies of autonomic control in athletes and nonathletes. This approach appears preferable to adding and subtracting parasympathetic and sympathetic influences to intrinsic HR, because the effects of direct parasympathetic and sympathetic nerve stimulations on HR are not simply additive.^{17,18} Our approach also provides a reasonable way to include central integration of the parasympathetic and sympathetic systems into the equations.

The assumption that the primary processes for parasympathetic reactivation and sympathetic withdrawal after first-order kinetics was based in part on principles of receptor kinetics. The time constant for parasympathetic reactivation would be analogous to an association constant and sympathetic withdrawal to a dissociation constant. The relation between the primary process (number of bound receptors) and the physiologic response (HR) is described as $1/(1 + P)$ for the parasympathetic system and $1 + (M \times S)/(M + S)$ for the sympathetic system. These simple mathematical descriptions were chosen empirically yet provided good results. It is possible that better mathematical functions could be derived for these relations. However, the key assumption underlying development of these equations is that it is possible to find a single descriptor that can reasonably represent a quantitative measure of dynamic parasympathetic function for a patient, with a similar descriptor for sympathetic function. The rate constants Tp and Ts serve this purpose. The differences between exercise levels are then attributed to the degree to which the sympathetic system is activated,¹⁹ with delayed long-term (>5 minutes) return to baseline after high levels of exercise attributable to resetting baseline sympathetic tone.

According to Rowell,²⁰ "the initial heart rate response to exercise, up to about $100 \text{ beats min}^{-1}$, is mainly attributable to the withdrawal of tonic vagal activity. . . . Above a heart rate of $100 \text{ beats min}^{-1}$ the slower sympathetic activation of the heart becomes the dominant factor in increasing rate." Our model is consistent with this proposal because all but 1 of our subjects attained HRs $>100 \text{ beats/min}$ (Figure 1), and setting $P_0 = 0$ did not alter our results. Further validation of our model could be obtained by correlating Tp and Ts with other measures of parasympathetic and sympathetic activities, but that is beyond the scope of this study.

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